response when injected subcutaneously, particularly if it is accompanied by an adjuvant.

C. Adjuvants

The response an immunogen is often enhanced if it is administered as a mixture with adjuvants. Adjuvants are substances that enhance the immunogenicity of an antigen. Example: Freund's incomplete antigen, Freund's complete antigen, Mycobacterium tuberculosis, Aluminum potassium sulphate (alum) and Bacterial lipopolysaccharide (LPS).

13.5.3 Epitopes

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Immune cells do not interact with or recognize an entire immunogen molecule instead; lymphocytes recognize discrete sites on the macromolecule called epitopes or antigenic determinants. Epitopes are the immunologically active regions of an immunogen that bind to antigen specific membrane receptors on lymphocytes or to secreted antibodies. Antigenic epitopes may consist of a single epitope or have varying number of the same epitope on the same molecule (Example: polysaccharides).

13.5.4 Haptens and the Study of Antigenicity

The pioneering work of Karl Landsteiner in the 1920s and 1930s created a simple, chemically defined system for studying the binding of an individual antibody to a unique epitope on a complex protein antigen. Landsteiner employed various haptens(small organic molecules that are antigenic but not immunogenic). Chemical coupling of a hapten to a large protein called a carrier, yields an immunogenic hapten-carrier conjugate.

13.5.5 Cross-Reactivity

When two antigens possess structurally similar antigenic determinants, the antibodies obtained to one of these antigens tend to react with the other antigen. These reactions are called cross reactions.

Infobits

Penicillin Allergy: New antigens are produced by altering epitopes. This can be done by conjugating haptens to the molecule. A classic example in human medicine is the **allergic response** of some persons to penicillin. A derivative of penicillin, **penicilloic acid** acting as a hapten, can couple with body protein and elicit an immune response that can be harmful, even life threatening, thus excluding this antibiotic from use in certain individuals.

13.6 Antibodies

The first real chemical information regarding the structure of antibodies was provided by Tiselius and Kabat in the early 1940s. They demonstrated that the gamma globulin fraction of serum proteins that migrated most slowly in electrophoresis contained most of the serum antibodies. This section deals with the structural and biological properties of antibodies (immunoglobulins).

Definition of antibodies

Antibodies are glycoproteins present in serum gamma globulins produced by

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